

Аппроксимация Паде

Баталов С.А.

In[605]:=

```
ClearAll["Global`*"]
```

Условие задачи

In[606]:=

```
Eq0 = y^4 - ε * y - 2 * y^2 - 8 == 0
```

Out[606]=

$$-8 - 2y^2 + y^4 - y\epsilon == 0$$

Решение при $\epsilon \rightarrow 0$

In[607]:=

```
Y0 = y[0] + ε * y[1] + ε^2 * y[2]
```

Out[607]=

$$y[0] + \epsilon y[1] + \epsilon^2 y[2]$$

In[608]:=

```
SolY0 = Solve[Table[Coefficient[Series[Eq0[[1]] /. {y -> Y0}, {ε, 0, 2}], ε, i] == 0, {i, 0, 2}], {y[0], y[1], y[2]}];  
Sol0 = (y -> Y0) /. SolY0
```

Out[609]=

$$\left\{ y \rightarrow -2 + \frac{\epsilon}{12} + \frac{5\epsilon^2}{1728}, y \rightarrow 2 + \frac{\epsilon}{12} - \frac{5\epsilon^2}{1728}, y \rightarrow -i\sqrt{2} - \frac{\epsilon}{12} - \frac{i\epsilon^2}{864\sqrt{2}}, y \rightarrow i\sqrt{2} - \frac{\epsilon}{12} + \frac{i\epsilon^2}{864\sqrt{2}} \right\}$$

Решение при $\epsilon \rightarrow \infty$

In[610]:=

```
EqInf = Collect[MultiplySides[Eq0 /. { $\epsilon \rightarrow 1 / \tau$ },  $\tau > 0$ ],  $\tau$ ]
```

Out[610]=

$$-y + (-8 - 2y^2 + y^4) \tau == 0$$

In[611]:=

```
YInf =  $\tau^k * x[1]$ ;
Collect[Expand[EqInf /. { $y \rightarrow YInf$ }], { $\tau$ ,  $\tau^k$ }]
```

Out[612]=

$$-8 \tau - \tau^k x[1] - 2 \tau^{1+2k} x[1]^2 + \tau^{1+4k} x[1]^4 == 0$$

In[613]:=

```
SolInf = { $y \rightarrow -8 / \epsilon$ ,  $y \rightarrow \epsilon^{1/3}$ }
```

Out[613]=

$$\left\{ y \rightarrow -\frac{8}{\epsilon}, y \rightarrow \epsilon^{1/3} \right\}$$

Переход к новому параметру $\mu = \epsilon^{1/3}$

In[614]:=

```
Sol0Mu = Sol0 /. { $\epsilon \rightarrow \mu^3$ }
```

Out[614]=

$$\left\{ y \rightarrow -2 + \frac{\mu^3}{12} + \frac{5\mu^6}{1728}, y \rightarrow 2 + \frac{\mu^3}{12} - \frac{5\mu^6}{1728}, y \rightarrow -i\sqrt{2} - \frac{\mu^3}{12} - \frac{i\mu^6}{864\sqrt{2}}, y \rightarrow i\sqrt{2} - \frac{\mu^3}{12} + \frac{i\mu^6}{864\sqrt{2}} \right\}$$

In[615]:=

```
SolInfMu = Simplify[SolInf /. {ϵ → μ^3}, μ > 0]
```

Out[615]=

$$\left\{ y \rightarrow -\frac{8}{\mu^3}, y \rightarrow \mu \right\}$$

Срачивание вещественных решений

Первый корень

In[616]:=

```
F1 = Sum[c[i] * μ^i, {i, 0, 3}] / Sum[d[i] * μ^i, {i, 0, 6}] /. {c[0] → 1}
```

Out[616]=

$$\frac{1 + \mu c[1] + \mu^2 c[2] + \mu^3 c[3]}{d[0] + \mu d[1] + \mu^2 d[2] + \mu^3 d[3] + \mu^4 d[4] + \mu^5 d[5] + \mu^6 d[6]}$$

In[617]:=

```
SolF1 = Solve[Union[Thread[CoefficientList[Numerator[F1] - Denominator[F1] * Sol0Mu[[1, 2]], μ] == 0][[1;9]],  
  {Coefficient[Series[Simplify[F1], {μ, Infinity, 3}], μ, -3] == -8}],  
  Union[Table[c[i], {i, 1, 3}], Table[d[i], {i, 0, 6}]]][[1]]
```

Out[617]=

$$\left\{ c[1] \rightarrow 0, c[2] \rightarrow 0, c[3] \rightarrow \frac{11}{720}, d[0] \rightarrow -\frac{1}{2}, d[1] \rightarrow 0, d[2] \rightarrow 0, d[3] \rightarrow -\frac{41}{1440}, d[4] \rightarrow 0, d[5] \rightarrow 0, d[6] \rightarrow -\frac{11}{5760} \right\}$$

In[618]:=

```
y1 = Simplify[(F1 /. SolF1) /. {μ → ϵ^(1/3)}]
```

Out[618]=

$$-\frac{8(720 + 11\epsilon)}{2880 + 164\epsilon + 11\epsilon^2}$$

Второй корень

In[619]:=

```
F2 = Sum[a[i] * μ^i, {i, 0, 5}] / Sum[b[i] * μ^i, {i, 0, 4}] /. {a[0] → 1}
```

Out[619]=

$$\frac{1 + \mu a[1] + \mu^2 a[2] + \mu^3 a[3] + \mu^4 a[4] + \mu^5 a[5]}{b[0] + \mu b[1] + \mu^2 b[2] + \mu^3 b[3] + \mu^4 b[4]}$$

In[620]:=

```
SolF2 = Solve[Union[Thread[CoefficientList[Numerator[F2] - Denominator[F2] * Sol0Mu[[2, 2]], μ] == 0][[1;;8]],  
Thread[CoefficientList[Series[Simplify[F2], {μ, Infinity, 0}], μ] == {0, 1}]], Union[Table[a[i], {i, 1, 5}], Table[b[i], {i, 0, 4}]]][[1]]
```

Out[620]=

$$\left\{ a[1] \rightarrow \frac{5}{22}, a[2] \rightarrow \frac{25}{264}, a[3] \rightarrow \frac{11}{144}, a[4] \rightarrow \frac{5}{288}, a[5] \rightarrow \frac{25}{6336}, b[0] \rightarrow \frac{1}{2}, b[1] \rightarrow \frac{5}{44}, b[2] \rightarrow \frac{25}{528}, b[3] \rightarrow \frac{5}{288}, b[4] \rightarrow \frac{25}{6336} \right\}$$

In[621]:=

```
y2 = Simplify[(F2 /. SolF2) /. {μ → ε^(1/3)}]
```

Out[621]=

$$\frac{6336 + 1440 \epsilon^{1/3} + 600 \epsilon^{2/3} + 484 \epsilon + 110 \epsilon^{4/3} + 25 \epsilon^{5/3}}{3168 + 720 \epsilon^{1/3} + 300 \epsilon^{2/3} + 110 \epsilon + 25 \epsilon^{4/3}}$$

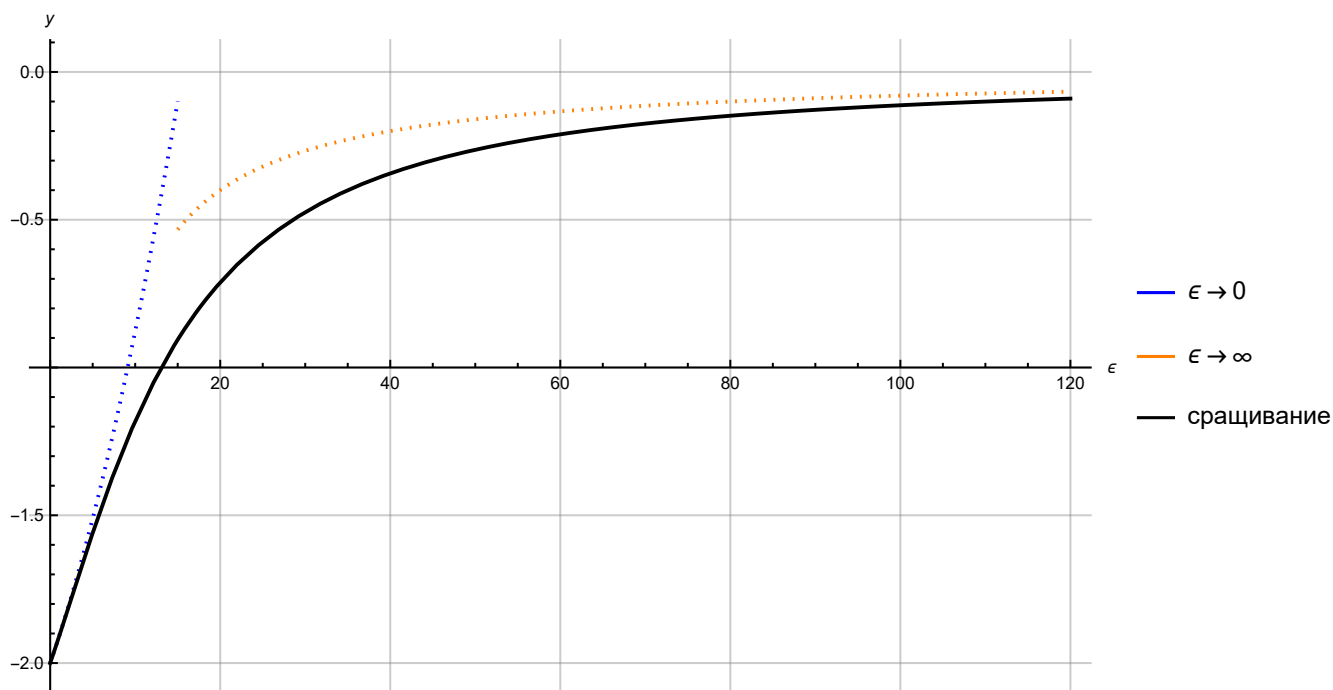
Построение решений

Первый корень

In[622]:=

```
Show[{Plot[Sol0[[1, 2]], {ϵ, 0, 15}, PlotStyle → {Blue, DotDashed}, PlotLegends → LineLegend[{Blue}, {"ϵ→0"}]],
      Plot[SolInf[[1, 2]], {ϵ, 15, 120}, PlotStyle → {Orange, DotDashed}, PlotLegends → LineLegend[{Orange}, {"ϵ→∞"}]],
      Plot[y1, {ϵ, 0, 120}, PlotStyle → Black, PlotLegends → LineLegend[{Black}, {"сращивание"}]],
      AxesLabel → {ϵ, y}, GridLines → Automatic, PlotRange → {Automatic, {-2, 0}}, AxesOrigin → {0, -1}, ImageSize → Large]
```

Out[622]=



Второй корень

In[623]:=

```
Show[{Plot[Sol0[[2, 2]], {ϵ, 0, 40}, PlotStyle → {Blue, DotDashed}, PlotLegends → LineLegend[{Blue}, {"ϵ→0"}]],
      Plot[SolInf[[2, 2]], {ϵ, 0, 120}, PlotStyle → {Orange, DotDashed}, PlotLegends → LineLegend[{Orange}, {"ϵ→∞"}]],
      Plot[y2, {ϵ, 0, 120}, PlotStyle → Black, PlotLegends → LineLegend[{Black}, {"сращивание"}]], AxesLabel → {ϵ, y},
      GridLines → Automatic, PlotRange → {Automatic, Automatic}, AxesOrigin → {0, 0}, ImageSize → Large]
```

Out[623]=

